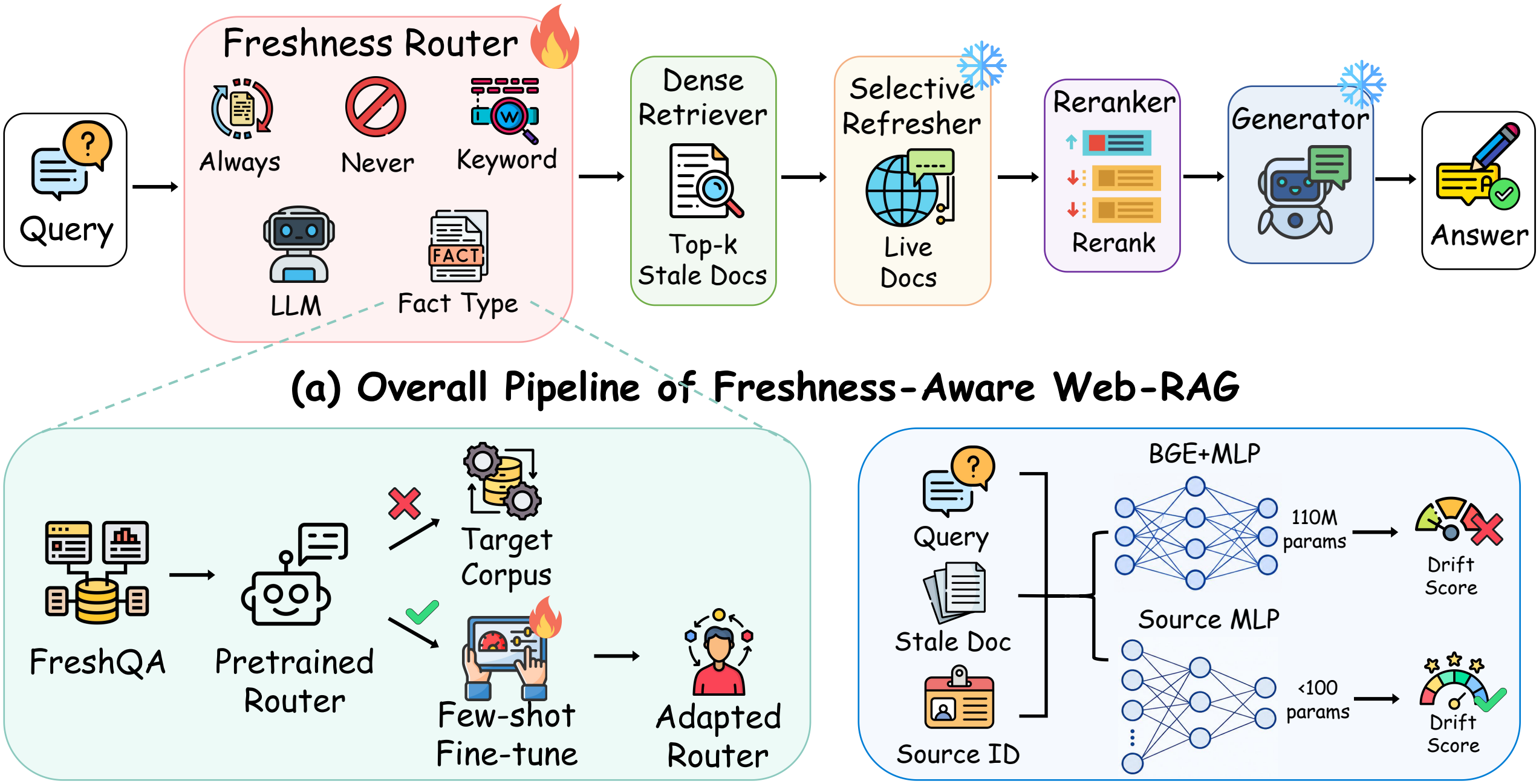

Freshness Router
Always
Never
Keyword
LLM
Fact Type
FACT
Query
Dense Retriever
Top-k Stale Docs
Selective Refresher
Live Docs
Reranker
Rerank
Generator
Answer
(a) Overall Pipeline of Freshness-Aware Web-RAG
FreshQA
Pretrained Router
Target Corpus
Few-shot Fine-tune
Adapted Router
(b) Cross-Corpus Few-shot Adaptation
Query
Stale Doc
Source ID
BGE+MLP
110M params
Drift Score
Source MLP
<100 params
Drift Score
(c) Drift Predictor